Yijie Ma

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Software engineer focused on distributed tracing, telemetry pipelines, and systems performance. gRPC maintainer with experience building OpenTelemetry instrumentation, Kubernetes networking, and AI-assisted engineering tools.

Skills

Languages: C/C++, Go, Python
Observability: OpenTelemetry, OTLP, Semantic Conventions, W3C Trace Context, Metrics, Distributed Tracing
Systems: Distributed Systems, gRPC, Kubernetes, API Gateway/L7 Proxy, Networking, Linux, Storage, RocksDB

Work Experience

Google — April 2017 - Present

Software Engineer

Google Cloud API Gateway & Service Infrastructure

- Led distributed tracing architecture for Gemini Agent Platform across HTTP/gRPC, defining tracing data models using OpenTelemetry semantic conventions for Cloud, GenAI, Model Context Protocol (MCP), and Agent-to-Agent (A2A) requests.
- Built W3C Trace Context propagation, span generation, sampling, and **asynchronous OTLP trace/log export pipelines** for API Gateway handling **millions of QPS**.
- Aligned Auth, Observability, and API Gateway teams on ambiguous agentic requirements; scoped milestones and mentored junior engineers through multi-quarter observability projects.
- Led delivery of an isolated fast API serving release track across control-plane and serving teams; pioneered reusable AI Skills to automate releases, adopted by other team members.

gRPC Open Source Remote Procedure Call Framework

- Core developer and maintainer of gRPC (40K+ GitHub stars), with **150+ authored PRs** and RFC/PR reviews across C++ core, networking, and observability.
- Co-authored gRFC A79 for non-per-call telemetry; architected the C++ Metrics framework and OpenTelemetry plugin, **reducing Counter overhead by 10%**.
- Built cross-platform asynchronous DNS resolution for Linux/POSIX and Windows; designed POSIX fork handling and resolved pre-fork deadlocks across gRPC C++ and Python.

Google Kubernetes Engine (GKE) Networking

- Built GKE control-plane automation in Go to provision multi-NIC RDMA network topologies for AI-optimized GPU clusters.
- Designed concurrent request handling with distributed transactions; validated multi-node GPU communication using NVIDIA NCCL tests.

Google Drive Desktop Application

- Built rsync-style differential uploads for millions of users; architected USB backup handling millions of files daily and **improved backup speed 10x** through I/O buffer optimization.
- Discovered a category of Windows crashes occurring **1,000+ times daily** in threads running on heap-allocated stacks.

Stadia Cloud Gaming Service

- Designed a Linux Cgroup library for VM container resource configuration.
- Used CPU profiling and container setup optimization to **cut game launch latency 75%** (1.58s to 396ms).

NetApp — August 2015 - March 2017

Member of Technical Staff

Designed filesystem software for AltaVault cloud-integrated storage to back up files to multiple cloud vendors; implemented full and incremental snapshot backup and restore using RocksDB.

Education

Carnegie Mellon University — Sep 2013 - May 2015

Master of Science in Electrical and Computer Engineering

Donghua University — Sep 2009 - May 2013

Bachelor of Engineering in Electrical Engineering and Automation